Maathrucare: An AI-Based Pregnancy Support System for Sri Lankan Mothers

INDIVIDUAL THESIS

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Individual component:

AI Chatbot with Multimodal Mood Detection (Facial Expression Recognition, Sinhala Text Sentiment and Mood Fusion

Group ID: R26-IT-146

Report submitted in partial fulfilment of the requirements for the

B.Sc. (Hons) Degree in Information Technology

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

[October 2026]

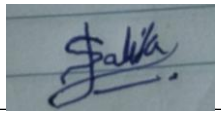
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ABSTRACT

Pregnancy brings emotional strain as well as physical change, yet antenatal care in Sri Lanka is built almost entirely around physical checks. Hospital studies report significant depressive or anxiety symptoms in roughly 16 to 23 percent of attending women, and the standard protocol includes no mental health screening. This thesis presents my component of Maathrucare, a pregnancy support application: a bilingual Sinhala and English chatbot that estimates a user's mood from her face and her messages, so that its support can adapt to how she is feeling.

I followed Design Science Research. For the facial signal, I fine-tuned MobileNetV2 on a de-duplicated FER-2013 training set (28,709 images reduced to 26,901) and evaluated it once on the held-out test split. Fine-tuning raised test macro-F1 from 0.4142 to 0.6036 (accuracy 0.6289). Temperature scaling reduced calibration error from about 0.30 to 0.013 without changing any prediction. Mapped to three application states (calm, neutral, distressed), the model reaches macro-F1 0.7681. That figure reflects an easier task, not a better model, and 24.3 percent of distressed faces were still missed. For text, I adapted SinBERT-small with a three-class head using 299 annotated Sinhala messages. It scored 0.6249 macro-F1 on validation but only 0.3620 on the locked test set, barely above chance, and it does not handle English. A weighted fusion layer combines the two signals and uses text alone when the camera is off. It passed all 15 conformance checks, but its accuracy could not be measured because no dataset pairs faces and messages with known mood labels.

The results show that a lightweight facial model is feasible, while Sinhala mood detection needs far more annotated data. My contribution is the facial emotion pipeline, the Sinhala sentiment model, the fusion design, and an honest evaluation of each.

Keywords: maternal mental health, facial expression recognition, Sinhala sentiment analysis, multimodal mood fusion, pregnancy chatbot

ACKNOWLEDGEMENTS

I would like to thank my supervisor, Dr. Kapila Dissanayake, for his guidance, feedback and patience throughout this research. His review of my experimental methodology, and in particular his approval of the annotation guideline for the Sinhala mood labels, shaped how carefully I handled the evaluation.

I am also grateful to my group members, IT22272386, IT22284716 and IT22557506, for working alongside me on Maathru care. Their components gave my chatbot something real to connect to.

I acknowledge the Department of Information Technology at the Sri Lanka Institute of Information Technology for the academic resources and facilities that supported this work. I also thank the creators of the FER-2013 dataset and the authors of the SinBERT model, whose openly shared work made my experiments possible.

Finally, I thank my family and friends .

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LIST OF ABBREVIATIONS

[Guidance - delete before submission: List abbreviations alphabetically. Define each abbreviation at first use in the main text even when it appears here.]

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
AUROC	Area Under the Receiver Operating Characteristic Curve
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
DSR	Design Science Research
ECE	Expected Calibration Error
EPDS	Edinburgh Postnatal Depression Scale
FER	Facial Expression Recognition
FHB	Family Health Bureau
GPU	Graphics Processing Unit
LLM	Large Language Model
MCH	Maternal and Child Health
ML	Machine Learning
MOH	Medical Officer of Health

Abbreviation	Meaning
NLP	Natural Language Processing
SDG	Sustainable Development Goal
SinBERT	Sinhala BERT-based language model
SLIIT	Sri Lanka Institute of Information Technology
TFLite	TensorFlow Lite
TRCSL	Telecommunications Regulatory Commission of Sri Lanka
UI	User Interface
UX	User Experience

1 INTRODUCTION

1.1 Overall project context

Sri Lanka has one of the stronger maternal health systems in South Asia. Antenatal care runs through the Family Health Bureau, government hospitals and the public health midwife network, and maternal mortality has fallen steadily over the past few decades. But the system is built around the body. A clinic visit checks blood pressure, hemoglobin, weight gain and the baby's position. It does not ask how the mother is coping, and psychological screening is absent from the national antenatal protocol at both primary and tertiary care levels [1]. Hospital studies suggest the cost of that omission: about 19.3 percent of women attending antenatal clinics in the Western Province showed significant depressive symptoms [2], and 22.8 percent showed anxiety in the third trimester [3]. Most of these women are never identified.

Maathrucare is the group's response to this gap. It is a mobile pregnancy support application with four components, each developed by a different member of the group (Group ID R26-IT-146):

- **Health monitoring and risk classification (IT22557506):** analyses heart rate, blood pressure and blood sugar, classifies risk, and alerts the doctor when risk is high.
- **CTG analysis (IT22272386):** analyses cardiotocography scans and values and produces draft clinical findings for doctor review.
- **Meal suggestions (IT22284716):** suggests Sri Lankan meals from the ingredients a user has at home, with nutritional analysis.
- **Chatbot with mood detection (IT22638168, this thesis):** a bilingual Sinhala and English chatbot that estimates the user's mood and adapts its support.

The first three components cover physical monitoring and nutrition. Mine covers the emotional side that clinic visits miss. It also sits between the patient and the doctor. In the proposed design, the mood history it produces is passed to the doctor-facing side of MaternaLink, so a doctor or midwife can see how a patient has been feeling between monthly visits.

The project is realistic in the Sri Lankan setting because smartphones are common. The TRCSL reported 67 percent smartphone use among adults aged 18 to 45 in 2024, with strong growth in rural provinces [4]. A mobile app can therefore reach women far from specialist services.

1.2 Individual component context

My component is the part of Maathrucare that talks to the user and tries to understand how she is feeling while it does. It has four parts that work together:

1. **A bilingual chatbot** that converses in Sinhala and English in a warm, informal register rather than a clinical one.
2. **A facial expression model.** The app finds the user's face in a camera frame and crops it. A MobileNetV2 classifier then returns probabilities for seven expressions: angry, disgust, fear, happy, neutral, sad and surprise.
3. **A Sinhala text model.** A SinBERT-small classifier reads a message and returns probabilities for three application mood states: calm, neutral and distressed.
4. **A fusion layer.** It maps the facial output to the same three states, combines it with the text output by weighted averaging, and returns one mood state. If the camera is off, the text result is used alone. If neither signal is usable, the layer returns "unknown" instead of guessing.

The chatbot is meant to use this mood state to adjust its tone and, when distress is detected, to suggest calming music.

I should be clear about what these mood states are. They are application states, not diagnoses. The system estimates how a user appears in a photo or how a message reads so the app can offer supportive content. It does not detect depression, anxiety or any clinical condition.

My work focused on the machine learning pipeline and its evaluation, which makes up most of this thesis: preparing and auditing the facial dataset, training and calibrating the facial model, building and testing the Sinhala sentiment model, and

designing and verifying the fusion layer. The group's mobile app, the other three components and any clinical use sit outside this thesis, except where my component connects to them.

1.3 Problem statement

Pregnant women in Sri Lanka meet the health system about once a month. In between, they cope alone, and the national antenatal protocol has no mental health screening to catch those who are not coping [1]. Stigma makes this worse. A woman is unlikely to tell a busy midwife that she is struggling unless someone asks. A digital companion available at any hour, in her own language, could fill the gap between visits. Building one raises three connected problems.

Language and context. Mental health chatbots are generally designed for English speakers and Western healthcare settings. A woman in a rural district who is most comfortable in Sinhala is unlikely to confide in an English-only app, and a generic app knows nothing about pregnancy. A chatbot for this group has to work in Sinhala, and Sinhala is still poorly served by language technology. Labelled datasets and pretrained models are scarce [5].

Unreliable single signals. A chatbot that waits for the user to say she is upset will miss the women who say nothing. Mood can be estimated passively from the face or from the text of a message, but each signal alone is weak. My own measurements show this. My facial model labelled 24.3 percent of genuinely distressed faces as neutral or calm, and it loses accuracy in dim light, which is common in homes. The Sinhala text model, trained on a small annotated set, performed only slightly better than chance on held-out data. Camera access is also optional, so a system that depends on the face fails whenever a user switches it off.

Combining the signals without good evidence. Facial and textual evidence have to be merged into one mood estimate, and the choice of how much to trust each signal changes the outcome. No dataset exists that pairs a person's face and message with a verified mood label, so the merged estimate cannot be validated the usual way. This is a methodological problem as well as an engineering one.

Together these problems lead to the central problem of this thesis: how can a bilingual pregnancy chatbot estimate a user's mood from facial expression and Sinhala text, honestly how reliable that estimate is, when the available models are weak and the data for validating them is scarce?

1.4 Research gap

The literature points to a gap at the point where three areas meet: maternal mental health in Sri Lanka, passive mood detection, and Sinhala language technology. Each area has been studied on its own, but I found little that connects them.

Gap 1: no Sinhala-language emotional support tool for pregnancy. Digital interventions for perinatal depression and anxiety have grown in number, but few have been properly evaluated, and apps for antenatal mental health in South and Southeast Asia are largely missing [6]. Sinhala adds a further barrier. It is under-resourced, with limited labelled data and few pretrained models [5]. I found no published chatbot that supports a Sinhala-speaking pregnant woman emotionally.

Gap 2: passive multimodal mood detection has not been applied to pregnancy support in this region .

Facial expression recognition, text sentiment analysis and multimodal fusion are each well studied [7]. The same review of perinatal apps found almost none that monitor mood passively [6]. I found no published work that combines a facial signal with Sinhala text in a pregnancy-specific setting.

Gap 3: honest evaluation of weak signals and missing modalities. Most work reports a single headline accuracy. For a wellbeing application, that is not enough, and my own results show why. Merging four expressions into one distress state raised my facial model's macro-F1 from 0.6036 to 0.7681 without improving the model at all. A model that reported confidence of 99 percent was right about 70 percent of the time until it was recalibrated. The Sinhala text model fell from 0.62 on validation to 0.36 on a locked test set. Systems also need a defined behaviour when the camera is off or the language is unsupported, and few published systems describe one. Fusion itself cannot be validated in the usual way because no dataset pairs faces and messages from the same person with a known mood label.

1.5 Aim and objectives

Aim

To design, build and evaluate the mood detection pipeline of a Sinhala and English pregnancy support chatbot, which estimates a user's mood from her facial expression and Sinhala text, combines the two signals, and reports honestly how reliable each part is.

Objectives

01. Build a facial expression model that is accurate enough to be useful and small enough for a phone.

Audit FER-2013 for duplicates and leakage, then fine-tune MobileNetV2 and evaluate it once on the held-out test split. The comparison point is a frozen-backbone baseline. The proposal's target was at least 65 percent accuracy, so I judge the result against that figure and against published FER-2013 results.

02. Make the facial model's confidence trustworthy.

Measure its calibration error and correct it with temperature scaling, without changing any prediction. This matters because the fusion layer weighs the model's confidence against the text signal.

03. Package the facial model for deployment.

Convert it to TensorFlow Lite in several precisions and compare size, accuracy and agreement with the original model. Measure speed on a real phone where possible.

04. Build a Sinhala text model that classifies messages as calm, neutral or distressed.

Annotate Sinhala pregnancy messages under a written guideline, adapt SinBERT-small to the three mood states, and test it once on a locked test set. The proposal's target was 70 percent accuracy.

05. Design a fusion layer that combines face and text evidence.

Map the seven facial expressions to the three mood states, combine the two signals

with a transparent weighted rule, keep working when the camera is off, and return "unknown" when no signal is usable. Verify the layer against its specification.

06. Evaluate each part honestly and state what the evidence does not show. Report the measured results, including those that miss their targets. Record the limits of the datasets, especially the Western bias of FER-2013 and the small size of the Sinhala data.

1.6 Research questions or hypotheses

01). Can a fine-tuned MobileNetV2, trained on de-duplicated FER-2013 data, recognise facial expressions well enough to be useful?

H1a: Test accuracy on the held-out split reaches at least 65 percent, the target set in my proposal.

H1b: Fine-tuning improves validation macro-F1 over a frozen-backbone baseline by more than the measured noise floor (0.0065).

02). Is the model's confidence reliable, and can it be corrected without retraining?

H2: Temperature scaling brings the Expected Calibration Error below 0.1, the usual boundary for poor calibration, without changing any prediction.

03). Does converting the model to TensorFlow Lite for deployment cost accuracy?

H3: At least one compressed variant stays within the noise floor on macro-F1 and agrees with the original model on nearly every individual prediction.

04). Can SinBERT-small, adapted on a small annotated set, classify Sinhala pregnancy messages into calm, neutral and distressed?

H4: Accuracy on the locked test set reaches at least 70 percent, the target in my proposal.

05). Does a weighted fusion layer behave correctly, and how much does the choice of weight matter?

H5a: The fusion layer follows its specification in every case, including a missing

camera, unusable text and the “unknown” outcome.

H5b: When the two signals disagree, the final mood state depends on the face weight.

06). Where does the evidence stop?

This one is a question without a numeric hypothesis. For each component, I ask what the measurements support, what they do not, and what data would be needed to settle the rest.

1.7 Scope limitations and assumptions

Scope

This thesis covers the machine learning pipeline behind the Maathrucare chatbot’s mood detection:

- Auditing and cleaning FER-2013, then fine-tuning, calibrating and evaluating a MobileNetV2 facial expression model.
- Converting that model to TensorFlow Lite and comparing the variants on size, accuracy and agreement with the original.
- Annotating Sinhala pregnancy messages under a written guideline, and adapting SinBERT-small to classify them as calm, neutral or distressed.
- Designing a fusion layer that combines the two signals, and checking it against its specification.

Limitations

- Dataset bias. FER-2013 is heavily Western, with few South Asian faces. Its images are small (48×48), grayscale, and contain label noise, with some byte-identical images filed under different emotions. I could not fix this within the dataset.
- Small Sinhala data. The annotated Sinhala set has 299 development records and the locked test set has 120 records, split across two languages. This supports a proof of concept, not a general performance claim.

- No fusion accuracy. No dataset pairs a person's face and message with a known mood, so fusion could be tested for correct behaviour but not for accuracy. Six fusion parameters, including the face weight, are not yet chosen.
- Phone speed unmeasured. I had no physical Android device connected, so inference latency on a phone was recorded as unmeasured. The latency figures I do have come from a development machine and should not be read as phone figures.

Assumptions

- The face is detected and cropped before it reaches the facial model, which was trained only on tight face crops.
- The three mood states are application constructs defined by my annotation guideline. They are not medical categories.
- The mapping from seven expressions to three states was fixed by my supervisor's judgement and then frozen. It is not derived from AffectNet valence data, because that data is not published in usable form.
- Calibration was fitted on validation data, and I assume it carries over to new data. Checking this would have needed a second use of the test set.
- Both test sets were used once, and I did no tuning after seeing the results.

1.8 Individual contribution

Research contributions

1. **A leakage audit of FER-2013.** I measured exact and near-duplicate images across all 35,887 files. About 5.2 percent were byte-for-byte duplicates, 557 duplicate groups crossed the train, validation and test splits, and 26 groups carried conflicting labels. I removed the cross-split duplicates from training only (28,709 images became 26,901) and left validation and test untouched so results stay comparable with published work.

2. **A measured noise floor.** I retrained configurations over three seeds and found a within-configuration standard deviation of 0.0065 macro-F1. I used this as the smallest difference I would call meaningful, so I could report some choices as "not distinguishable" instead of calling them wins.
3. **A facial expression model with an honest evaluation.** Fine-tuning MobileNetV2 raised test macro-F1 from 0.4142 to 0.6036. The test set was used once. The generalisation gap was small (−0.0085 macro-F1), and I tested whether the model was exploiting image brightness or watermarks (it was not).
4. **A calibration fix.** The model was badly overconfident (ECE 0.30). Temperature scaling (T = 5.727) brought it to 0.013 with no prediction changed, so every accuracy figure stayed valid.
5. **A deployment comparison.** I compared four TensorFlow Lite variants, rejected full-integer quantization (it lost 0.1188 macro-F1), and found that per-image agreement can hide differences that macro-F1 does not show.
6. **A Sinhala mood dataset and model.** I worked with an annotation guideline, built a 299-record development set, adapted SinBERT-small, and evaluated it once on a locked 120-record test set. The result was weak (macro-F1 0.3620). I also found that the model silently predicts "calm" for English text.
7. **A fusion design with a sensitivity analysis.** When face and text disagree, the final state flips only when the face weight falls between 0.50 and 0.60, so an "equal weights" default is not a safe placeholder.

Product contributions

- A facial expression service that returns calibrated probabilities for the seven classes, with a written input and output contract.
- A Sinhala sentiment service, verified to reproduce the recorded results (76 of 76 predictions matched, with no difference through the web layer).

- A fusion layer that handles a missing camera, discards unusable text, returns "unknown" when it has no evidence, and refuses to run without explicit parameters. It passes 15 conformance checks.

What I did not contribute

I did not build the other three Maathrucare components. I have no accuracy figure for fusion. I have no phone latency measurement.

1.9 Thesis structure

This thesis has eight chapters, followed by the references, a glossary and two appendices.

Chapter 1, Introduction, has set out the context of the Maathrucare project, my individual component, the problem, the research gap, the aim and objectives, the research questions, the scope and my contribution.

Chapter 2, Literature review and related work, explains how I searched the literature and reviews the background on antenatal mental health, facial expression recognition, Sinhala language technology and multimodal fusion. It then examines existing apps and ends by showing where the gap lies.

Chapter 3, Research methodology, describes the Design Science Research approach, the datasets, the experimental procedure and the evaluation metrics. It also covers how I handled the test sets, and the ethics, privacy and responsible AI decisions.

Chapter 4, Product and solution design, sets out the requirements for my component and how it fits into the complete Maathrucare system. It discusses the design alternatives I considered and gives the detailed design of the facial service, the sentiment service and the fusion layer.

Chapter 5, Implementation, integration and testing, covers the development environment, how each part was built, how the parts connect, and the verification tests I ran. It also states what has and has not been deployed.

Chapter 6, Results and validation, reports the measured results for the facial model, calibration, the TensorFlow Lite variants, the Sinhala sentiment model and the fusion layer. It says plainly which hypotheses were supported.

Chapter 7, Discussion, interprets the findings and compares them with published work. It sets out my research and product contributions and discusses the limitations and threats to validity.

Chapter 8, Conclusion and future directions, summarises what was achieved against each objective and proposes the next steps. These include English sentiment, a paired face and text dataset, a study with real users and a phone benchmark.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

I reviewed the literature in five areas that match the five parts of my component: antenatal mental health in Sri Lanka, facial expression recognition, Sinhala language technology, multimodal mood fusion, and digital support tools for pregnancy. I did not set out to cover all of affective computing. Where a topic was far from the decisions I had to make, I left it out.

Sources. I searched Google Scholar, IEEE Xplore, PubMed and the ACM Digital Library. I also searched Sri Lankan sources directly: the Ceylon Medical Journal, the Sri Lanka Journal of Obstetrics and Gynecology, and reports from the Family Health Bureau, the Department of Census and Statistics and the TRCSL. For the models I used, I read the original papers and the dataset and model documentation.

Search terms. I combined terms such as *antenatal depression Sri Lanka*, *perinatal mental health chatbot*, *facial expression recognition FER-2013*, *Sinhala sentiment analysis*, *SinBERT*, *multimodal emotion fusion* and *model calibration*.

Inclusion and exclusion. I kept peer-reviewed papers, official statistics and original dataset or model papers, mostly from 2025 onwards. I made exceptions for foundational work such as MobileNetV2. I kept studies set in Sri Lanka whenever they existed, since local prevalence and language matter here. I excluded blog posts and app-store descriptions as evidence of effectiveness. I used company websites only to describe what a product claims to do, and I say so wherever I do this.

How I used the sources. I used the literature for three things: to establish the problem (Chapter 1), to choose methods (Chapters 3 and 4), and to compare my results with published ones (Chapter 7.2). Published scores are not directly comparable with mine in every case. FER-2013 scores, for example, are reported with different protocols, so I only compare with work that uses the standard split.

Limits of this review. It is a targeted review, not a systematic one. I did not follow a formal protocol such as PRISMA, and I did not register a search plan. I read only English-language sources. Sinhala-language literature may exist that I could not read.

For these reasons, when I say a combination of methods has not been published, it means I did not find one, not that none exists.

2.2 Domain and theoretical foundations

2.2.1 Mood in pregnancy and how it is measured

Pregnancy is a period of raised risk for depression and anxiety, and the consequences reach the baby. A meta-analysis of prospective studies linked depression during pregnancy to a higher risk of preterm birth and low birth weight [8]. This is why the missing screening described in Chapter 1 matters clinically as well as for the mother's own wellbeing.

The usual screening tool is a short self-report questionnaire. The Edinburgh Postnatal Depression Scale (EPDS) has ten items and was designed for the postnatal period, though it is also widely used during pregnancy [9]. The Sri Lankan prevalence studies in Chapter 1 relied on this kind of screening. A questionnaire needs the woman to complete it, and it captures one moment. My component works from a different premise: estimating mood continuously from signals she produces anyway, without asking her to fill anything in. That makes it a complement to screening, not a substitute for it.

2.2.2 Representing emotion

Facial expression datasets label images with discrete categories. The seven FER-2013 labels (angry, disgust, fear, happy, neutral, sad, surprise) follow the "basic emotions" tradition, which holds that a small set of expressions is recognised across cultures [10]. The alternative is a dimensional view such as the circumplex model, which places emotions on two axes: valence (pleasant to unpleasant) and arousal (calm to excited) [11]. The two views matter for my work in different ways. The categories are what the face model outputs. The dimensions are the standard argument for deciding which categories belong together when collapsing seven expressions into three application states, for example that surprise is neutral in valence while anger and fear are negative.

There is a deeper caution. A review of the evidence argued that facial movements do not reliably map to specific inner emotions, and that expressions vary with context and culture [12]. For this reason I describe the system as estimating how a user *appears*, not what she feels. It supports the decision in Chapter 1 to treat my three states as application constructs and not as medical categories.

2.2.3 Facial expression recognition

FER-2013 comes from a 2013 machine learning challenge and holds 35,887 grayscale images of 48×48 pixels in seven classes [13]. Its authors noted human accuracy of around 65 percent on it, which reflects noisy labels and ambiguous faces. A later effort relabelled the images using multiple crowd annotators to reduce this noise [14], which shows how much label quality limits the task. AffectNet is a much larger collection of web images with both categorical labels and valence and arousal values [15]. I trained only on FER-2013 and explain the reasons in Chapter 3.

Convolutional networks dominate this field. For a mobile application, size and speed matter as much as accuracy. MobileNetV2 was designed for this setting, using depthwise separable convolutions with inverted residual blocks to keep the parameter count low [16]. I chose it for that reason and used transfer learning, starting from ImageNet weights, since the training set is small.

2.2.4 Calibration

A classifier's softmax output looks like a probability, but modern neural networks are often overconfident, meaning their stated confidence exceeds their accuracy [17]. The standard measure is Expected Calibration Error (ECE), the average gap between confidence and accuracy across confidence bins. Temperature scaling is a simple fix. It divides the network's raw outputs by a single learned value and does not change which class wins [17]. Calibration matters for fusion, because a weighted average is only meaningful if each signal's numbers mean what they claim to.

2.2.5 Sinhala language technology

Sinhala is spoken by most Sri Lankans, but it is under-resourced in labelled data and pretrained models [5]. SinBERT is a transformer pretrained on Sinhala text and has been reported to outperform multilingual models on Sinhala classification tasks [18].

Those results come from benchmarks such as news comments, not from conversations about pregnancy. A good score there does not predict a good score on my task, and my results in Chapter 6 bear this out.

2.2.6 Multimodal fusion

When several signals describe the same thing, they can be combined at different stages. Feature-level fusion merges the raw inputs before classification. Decision-level fusion lets each model decide first and then combines the outputs [7]. Surveys of multimodal learning treat fusion as one of several core challenges and note the problem of missing modalities [19]. I used decision-level fusion with a weighted average. It is easy to inspect, it allows each model to be trained separately on its own data, and it degrades cleanly when one signal is absent.

2.3 Related research

Antenatal mental health in Sri Lanka. Hospital-based studies found significant depressive symptoms in about 19 percent of attending women in the Western Province [2] and third-trimester anxiety in about 23 percent at one Colombo hospital [3]. A policy analysis confirmed that psychological screening is absent from the national antenatal protocol and suggested mobile technology as a practical way to deliver support [1]. These sources establish the need. They do not describe any tool.

Digital support for perinatal mental health. A review of digital interventions for perinatal depression and anxiety reported that few had been properly evaluated and that apps for South and Southeast Asia were largely missing [6]. This is the closest published statement of my gap.

Conversational agents. A randomised trial of the Woebot chatbot in young adults found reduced depressive symptoms over two weeks compared with an information-only control [20]. It shows that a chatbot can deliver benefit, but the participants were not pregnant, the interface was English and the agent did not use the user's face or other passive signals.

Facial expression recognition. Work on FER-2013 typically reports accuracy in the 65 to 73 percent range, with human accuracy near 65 percent [13]. Mobile-oriented

work uses compact networks like MobileNetV2 [16]. My model's results are compared with this range in Chapter 7.2.

Multimodal affect. Reviews of multimodal emotion recognition conclude that combining modalities usually helps, because each fails in different conditions [7], [19]. Most of this work uses curated datasets in which several modalities are recorded together from the same person. That condition does not hold in my project, which is a point I return to in 2.5.

Sinhala sentiment. Sinhala-specific encoders like SinBERT improve on multilingual baselines for Sinhala text [18], but I found no Sinhala resource labelled for mood in a pregnancy setting. That is why I had to build a small annotated dataset of my own.

2.4 Existing systems products and techniques

Existing option	What it offers	Where it falls short for this use
Public antenatal care (FHB, MOH clinics)	Monthly physical checks, midwife visits, printed health booklets	No emotional screening and no contact between visits
General mental health chatbots	Conversational support based on cognitive behavioral techniques [20]	Not pregnancy-specific, English-focused, and I found no passive mood sensing
General pregnancy apps (for example Glow, BabyCenter)	Tracking of milestones and fetal development information	Information tools with little emotional support
Mood diary apps	A place to record how one feels	Relies on the user to log her mood, and does not respond to what she enters

2.5 Gap synthesis and conceptual framework

First, the need is well documented in Sri Lanka, and the policy literature already suggests mobile delivery [1]. Second, the building blocks exist separately: compact facial models [16], Sinhala encoders [18] and fusion methods [7], [19]. Third, I found no published system that joins them for Sinhala-speaking pregnant women, and little published guidance on evaluating such a system honestly when the data is thin. Fusion in particular is usually assessed on datasets that record several modalities from one person, and no such dataset exists here.

My conceptual framework follows from this. A user produces two signals, a face image and a text message. A face detector crops the face, and the facial model returns seven expression probabilities, which are calibrated so that they mean what they say. A mapping turns these into three application states. The Sinhala text model returns the same three states directly. The fusion layer weighs the two, uses whichever is available when one is missing, and returns “unknown” when neither is usable. The chatbot then adjusts its support according to the result. Each box in this chain is evaluated on its own, and the results are reported with their limits stated.

3 RESEARCH METHODOLOGY

3.1 Research design

I used Design Science Research (DSR), because the aim was to build something and then judge it against stated criteria, not only to study a phenomenon [21]. DSR works well for a project like this one because it treats the built artefacts, here a facial model, a text model and a fusion layer, as the research output, and it expects them to be evaluated and refined [22].

The work followed one repeating cycle, applied separately to each artefact:

1. Define what the artefact must do and how I will measure it, before running anything.
2. Build a baseline.
3. Change one factor at a time and keep a change only when it beats the baseline by more than the measured noise.
4. Evaluate the chosen design once on a held-out set.
5. Record the result and the limits of what it shows.

Three rules ran through all of it. Selection decisions used validation data only. Each held-out test set was used once. And I recorded results that missed their targets, because a method that only reports successes cannot be trusted.

The research is quantitative. For the face model, I compare configurations on macro-F1. For the text model, I compare against the proposal's accuracy target and a chance baseline. For fusion, I test whether the layer behaves as specified, because I cannot measure its accuracy

3.2 Data participants or study materials

No pregnant women or other participants took part in this research. I worked with two datasets, which I describe separately.

3.2.1 Facial expression data

I used FER-2013, which holds 35,887 grayscale face images of 48×48 pixels in seven classes (angry, disgust, fear, happy, neutral, sad, surprise) [13]. I followed its canonical split so my results stay comparable with published work:

Split	**Source**	**Images**
Train	Training set	28,709
Validation	PublicTest	3,589
Test	PrivateTest	3,589

able 3.1: FER-2013 split used in this thesis.

An audit before training found that 5.2 percent of files were exact duplicates, that duplicates crossed the train, validation and test splits, and that some identical images carried different labels. I removed the cross-split duplicates and within-train redundancy, plus 53 images flagged as anomalies (for example, all-black images), from the training set only. This reduced it from 28,709 to 26,901 images. I left validation and test unchanged. I recorded every cleaning decision in manifest files and deleted nothing from the raw data, so each decision can be reversed.

I also tested a face detector (BlazeFace) as a cleaning filter and rejected it. It found a face in 99.94 percent of images, which suits it as a deployment detector, but its confidence tracked pose and occlusion rather than image quality, so using it to filter would have removed real but difficult faces. I explain this in Chapter 4.

I chose FER-2013 alone. My proposal planned to add AffectNet, but I did not use it.

3.2.2 Sinhala text data

The text data is a small set of Sinhala and English pregnancy-related messages, each given one of three application mood labels:

- **CALM:** relief, reassurance, contentment or settled coping is dominant.
- **NEUTRAL:** the content is factual or its emotion is unclear, or no emotion is clearly dominant.

- **DISTRESSED:** fear, worry, sadness, anger, overwhelm or loss of control is expressed or reasonably implied.

These labels are defined by a written annotation guideline (version 2.1), approved by my supervisor on 29 August 2026. It defines mood as an application construct for a single message, not as sentiment or clinical severity. A pregnancy topic or a negative word alone does not make a message distressed.

Annotation followed these rules. Two annotators labelled each message independently, without seeing each other's labels or any model output. Disagreements and ambiguous cases went to an adjudicator, who wrote the final label. Original annotations were kept. A record the adjudicator could not defensibly label was marked UNRESOLVED and excluded, never relabelled.

The data was organised as follows:

Set	**Records**	**Use**
Annotated source	300 (English and Sinhala)	Origin of the locked test set and of 90 historical development records
Locked test set	120 (20 per class for each language)	One final evaluation
Development set (Dev-v2), Sinhala only	299 (CALM 70, NEUTRAL 183, DISTRESSED 46)	Training and validation

Table 3.2: Sinhala and English text data.

The development set combines 90 earlier non-frozen Sinhala records and 209 newly annotated records. Thirty-one new records were marked UNRESOLVED and left out. I split the development set 75/25, stratified by class, with a fixed seed (20260828), giving 223 training and 76 validation records. Checks found no overlap between development records and the locked test set. Raw annotator agreement on the earlier non-frozen records was 91.67 percent.

The mood classes are unbalanced, with NEUTRAL making up about 61 percent of the development set. I did not rebalance the data, and used class weights during training.

3.3 Experimental or study procedure

Step	What I did
1. Dataset audit	Measured duplicates, leakage and image quality
2. Cleaning	Applied the cleaning policy to the training set only
3. Frozen baseline	Trained a new classifier head on a frozen ImageNet MobileNetV2 and compared two input sizes (96 and 128 pixels) and two class-weighting options
4. Noise floor	Repeated the baseline over three seeds (42, 1337, 2024) to measure run-to-run variation
5. Fine-tuning	Searched one factor at a time: augmentation, how many layers to unfreeze, batch normalisation, class weighting
6. Final evaluation	Evaluated the frozen baseline and the fine-tuned model once on the test set
7. Error analysis	Studied confusion patterns, lighting and pose effects, brightness shortcuts and Grad-CAM attention maps (validation images only)
8. Calibration	Fitted temperature scaling on validation
9. Conversion	Exported four TensorFlow Lite variants and compared them on validation
10. Benchmark	Measured model size and memory, and wrote a protocol for on-device timing

Table 3.3: Facial model experimental sequence.

Fine-tuning started from the trained baseline head, not from a random one, to avoid large early gradients damaging the pretrained features. The fine-tuning settings were

an Adam optimiser with learning rate 1e-4, batch size 64, at most 40 epochs, early stopping on validation macro-F1 with patience 8, and learning-rate reduction on plateau. Input size was 96×96.

3.3.2 Sinhala text model

I adapted SinBERT-small (pinned to a fixed model revision) by replacing its sentiment head with a new three-class head for CALM, NEUTRAL and DISTRESSED. I ran two development experiments.

- **Experiment 01** used 90 Sinhala records (72 train, 18 validation).
- **Experiment 02** used the 299-record Dev-v2 set (223 train, 76 validation).

Settings for Experiment 02 were AdamW, learning rate 2e-5, weight decay 0.01, batch size 8, maximum length 512 tokens, at most 5 epochs, and class-weighted cross-entropy with weights computed from the training fold only. I selected the checkpoint on validation macro-F1 with early stopping (patience 2). Training ran on a CPU.

The chosen checkpoint then received one authorised evaluation on the locked 120-record test set. Before that run I confirmed that the file's identifiers matched the locked manifest. I did no tuning afterwards.

3.3.3 Fusion layer

I designed the fusion layer to a written specification and tested it against that specification, because I could not test its accuracy. The steps were:

1. Map the seven facial probabilities to three states (see 3.5 and Chapter 4).
2. Combine the facial and text scores by weighted average.
3. Define behaviour for each availability case: both signals, face only, text only, neither.
4. Run 15 conformance checks covering the combination rule, the availability table, the "unknown" outcome and a refusal to run without explicit parameters.

5. Run a sensitivity sweep of the face weight from 0 to 1 over eight constructed scenarios, to see where the weight changes the outcome.

3.4 Measures and evaluation metrics

Component	Metric
Face model	**Macro-F1** (primary), accuracy, per-class precision, recall and F1, confusion matrix
Face model	**Noise floor** (0.0065 macro-F1)
Face model	**Expected Calibration Error (ECE)**, reliability bins, AUROC of confidence as a correctness detector
TensorFlow Lite variants	File size, memory proxy, macro-F1, ECE, per-sample **argmax agreement** with the original model
Sinhala model	Accuracy, macro-F1, weighted F1, per-class results, confusion matrix
Sinhala model	Chance level (macro-F1 about 0.333 for three classes)
Fusion	Conformance checks passed, weight sensitivity

Table 3.4: Evaluation measures.

The noise floor works like this. I trained the same configuration with three seeds and took the average standard deviation of validation macro-F1. A difference smaller than that is reported as “not distinguishable”, not as an improvement.

The proposal listed further targets: inference time under 3 seconds per frame, 70 percent agreement with users’ self-reported mood, a satisfaction score of 4 out of 5, and 100 percent offline access to history and music. I did not measure these, because they need a phone, users or features outside this thesis. I report them as not evaluated in Chapter 6.

3.5 Data analysis

Comparing configurations. I compared facial model configurations on validation macro-F1 and used the noise floor as the threshold for a meaningful gain. Final selection used three-seed means. The test set was then used once for the frozen baseline and the fine-tuned model together.

Generalization. I checked that selection on validation had not overfit by comparing validation and test macro-F1 for the same model.

Error analysis. I computed row-normalized confusion matrices, measured accuracy by brightness and by face-detection confidence, tested whether errors fall closer to the predicted class's mean brightness (a shortcut check), and used Grad-CAM at the last feature map to see where the model attends. I ran these on validation images or on predictions already exported, so the test set stayed untouched after its one use.

Mapping seven expressions to three states. The application needs three states, so I mapped the seven classes as follows: happy to CALM; neutral and surprise to NEUTRAL; angry, disgust, fear and sad to DISTRESSED. My supervisor settled the two disputed classes, surprise and disgust, by expert judgement, consistent with the circumplex model [11]. I could not use AffectNet valence and arousal averages because they are not published per class, so I do not describe the mapping as derived from AffectNet. I froze the mapping before the three-state test evaluation so that the reported result describes a fixed configuration.

Two outputs come from this mapping and they serve different purposes. For a standalone prediction, I take the most probable expression and translate that label (I call this Rule A). For fusion, which multiplies numbers by weights, I use grouped probability sums (calm = happy; neutral = neutral + surprise; distressed = angry + disgust + fear + sad). When only the face is available, the system always returns the Rule A label.

Why fusion has no accuracy score. Measuring fusion accuracy needs data where one person's face and message, at the same moment, carry a verified mood label. No such dataset exists. FER-2013 has faces with expression labels, and my Sinhala data

has messages with mood labels, and there is no valid way to join them. Pairing unrelated records would invent a relationship that does not exist, so I did not do it.

3.6 Ethics privacy security and responsible AI

Participants and consent. No human participants took part, so no consent forms or ethics clearance were needed for this thesis. My proposal planned usability sessions with adult volunteers under SLIIT ethical review, but I did not run them. Any future user study must go through that review first.

Training data. FER-2013 is a public research dataset. I collected no facial images from any person. Face plots that could show identifiable people were not committed to the repository, and the rule for including such figures in this thesis remains with my supervisor.

Annotator and message data. The Sinhala messages were de-identified before annotation, and the quality checks included a de-identification check. The annotators labelled messages under a written guideline that records its version and hash.

Live user privacy. This needs a careful statement, because my design changed after the proposal. The proposal said facial analysis would run entirely on the phone, with no image ever leaving it. My final design has a facial inference service that receives a cropped face by upload. The service's contract says it does not store images, and its input rules reject URLs because a URL would imply the image had been stored somewhere first. I describe the privacy guarantees of the final architecture accurately in Chapters 4 and 5, and I do not repeat the proposal's on-device claim.

Responsible AI.

- No diagnosis. The three states are application constructs. The system does not detect depression or any clinical condition. It estimates how a person appears and how a message reads.
- *Bias.* FER-2013 is heavily Western, with few South Asian faces. I did not infer any demographic attribute from the images, because doing so would be

unsound and ethically inappropriate. I state this as a limitation, not a solved problem.

- Missing evidence. The system never invents a face signal when the camera is off. It uses text alone, and returns “unknown” when neither signal is usable.
- Language hazard. The text model handles Sinhala only and returns a confident “calm” for English text. Fusion discards text evidence that is not Sinhala, which requires language detection that the backend must supply.
- Honest confidence. I calibrated the face model before using its confidence for anything.

3.7 Reproducibility and research artefacts

I wrote each run’s metadata to its own record, with the seed, package versions, dataset SHA-256 hash, hyperparameters and metrics. A result without such a record is not counted as a result. My notebooks reproduced the first notebook’s headline figures exactly after I re-ran them in a different environment with different library versions, which confirmed the audit numbers.

Item	Detail
Facial training environment	WSL2 Ubuntu, NVIDIA RTX 3050 6 GB laptop GPU, TensorFlow 2.21.0, Keras 3.15.1, mixed precision with a float32 output layer
Dataset audit environment	Conda environment with Python 3.10, MediaPipe and OpenCV
Sinhala training	CPU only
Seeds	42, 1337, 2024 for the face model; 20260828 for the Sinhala split
Dataset pinning	SHA-256 hashes for the FER manifests, the Dev-v2 release, the split, the frozen

	test file and the final checkpoint
Cleaning	Expressed as manifest files (canonical, cleaned and image-flag lists) so every decision can be reversed
Model versions	SinBERT-small pinned to a fixed revision; TensorFlow Lite variants exported from the validated model

Table 3.5: Reproducibility details.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Stakeholders

Stakeholder	What they need from this component
Pregnant women (primary users)	Support in Sinhala that responds to how they are feeling, without having to fill in forms, and a text-only mode if they switch the camera off
Doctors and midwives	A summary of mood between clinic visits (proposed, see 4.2)
MaternaLink backend developer	A clear contract for calling the models, without needing to read the training notebooks
Researchers and examiners	Results that can be reproduced and claims that match the evidence

4.1.2 Functional requirements

ID	Requirement	Status
FR1	Estimate seven facial expression probabilities from a cropped face	Met (model and service)
FR2	Return calibrated probabilities with the model version and a hash of the model file	Met
FR3	Classify a Sinhala message as calm, neutral or distressed	Met, with weak accuracy (Chapter 6)
FR4	Combine face and text evidence into one mood state	Built and conformance-tested, accuracy not

		measured
FR5	Continue with text alone when the camera is off	Met in fusion
FR6	Return “unknown” when no usable evidence exists, instead of guessing	Met in fusion
FR7	Reject text that is not Sinhala before it reaches fusion	Met in fusion; the language detector itself is not built
FR8	Never store or log user text or face images	Tested for text; image handling is by design
FR9	Let the chatbot adjust its tone and suggest support when distress is detected	Deferred (chatbot is not part of the ML work)
FR10	Provide chat history and music offline	Deferred
FR11	Pass a mood summary to the doctor-facing side	Deferred
FR12	Support English sentiment	Out of scope (future work)

Table 4.1: Functional requirements and their status.

4.1.3 Non-functional requirements

ID	Requirement	Status
NFR1	**Privacy:** no face image or user text is stored or logged by the services	By design; tested for text
NFR2	**Honesty of output:** confidence values must match real accuracy	Met for the face model after calibration (ECE 0.0126)
NFR3	**Traceability:** every prediction records which model produced it	Met (model version and SHA-256 hash)

NFR4	**Fail loudly:** a service must refuse to run if it loads the wrong model	Met (contract check)
NFR5	**Never block the chat:** if the face service is slow or down, the conversation continues without it	Specified in the integration guide
NFR6	**Small model size** for possible phone use	Met (2.5 to 9 MB variants)
NFR7	**Inference time under 3 seconds per frame on a mid-range phone**	Not measured (no device)
NFR8	**Sinhala Unicode** must survive transmission unchanged	Met (Sinhala service output matched direct output exactly)
NFR9	**Reproducibility:** results can be regenerated from recorded configurations	

Table 4.2: Non-functional requirements and their status.

4.2 Component architecture within the complete solution

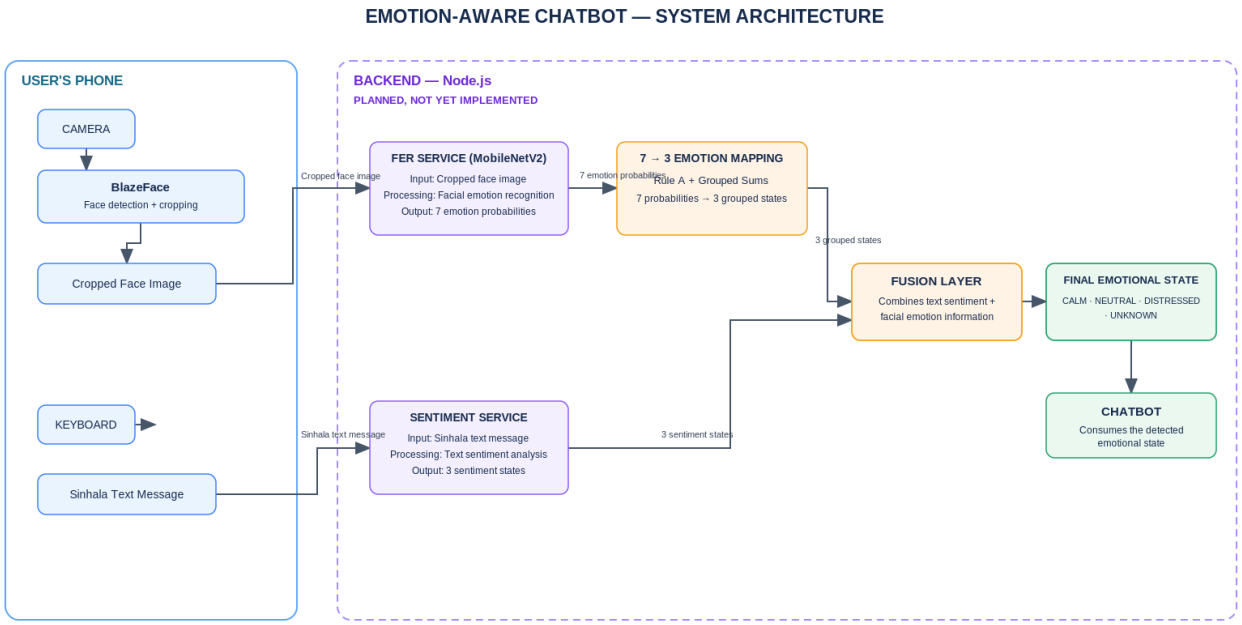


Figure 4.1: Component architecture.

Part	What it does	Where it runs	Status
BlazeFace	Finds the face and crops it	User's phone	Validated on FER-2013 (99.94% detection)
FER service	Face crop → seven calibrated probabilities	Server (planned Hugging Face Space)	Built and tested locally, not deployed
Sentiment service	Sinhala text → three mood probabilities	Server	Built and tested locally, not deployed
Fusion layer	Combines the two signals	Server	Built and conformance-tested
Backend	Calls the services and stores mood	Server	Not built

	evidence		
Chatbot	Uses the mood state to adapt its replies	App and backend	Not part of this thesis's ML work

Table 4.3: Parts of the component, where they run and their status.

Three design points follow from this arrangement:

- **The face detector is a separate step.** The facial model was trained only on tight face crops. Given a whole photo, it still returns seven plausible numbers, and nothing in the response shows they are meaningless. Detection therefore happens upstream and cropping is the caller's job.
- **Face and text are processed independently.** Neither service knows about the other, and only the fusion layer sees both. This lets each be trained, tested and replaced on its own.
- **Mood sensing is auxiliary.** If the face service is slow or down, the conversation continues without it. Chat responses must never wait on it.

4.3 Design alternatives and decisions

Decision	Alternatives I considered	Why I chose this
Seven-class facial model, mapped to three states afterwards	Train a three-class model directly (my proposal)	Keeps the mapping changeable without retraining. The mapping was still undecided when the model was trained
MobileNetV2 with ImageNet transfer learning	Training from scratch, EfficientNet-Lite (proposal fallback)	Small, built for mobile, and fine-tuning needs less data than training from scratch
Fine-tuning, not a frozen backbone	Frozen backbone only	Frozen features were the ceiling (validation macro-

		F1 0.4357 against 0.6063 fine-tuned)
Input 96×96	128×128, or native 48×48	128 was not better than 96 by more than the noise floor (+0.0022), so I took the cheaper size
Unweighted classes	Balanced class weights	Balanced weights rescued `disgust` recall but cost more elsewhere (+0.0220 macro-F1 for unweighted)
Temperature scaling before using confidence	Using raw softmax; Platt scaling	The raw model was badly overconfident (ECE 0.30). One scalar fixes it and cannot change any prediction
BlazeFace as detector, not as data filter	Using its confidence to clean FER-2013	Its confidence tracks pose and occlusion, so filtering would delete real but hard faces. It would remove 43 junk images and 818 real ones at a 0.45 threshold
Reject full-integer int8	Use it for speed	Lost 0.1188 macro-F1 (18× the noise floor), wrecked calibration, and was larger than dynamic int8
float32 on the server	float16 (recommended in notebook 07), dynamic int8	Server hardware is x86, where float32 was most accurate and fastest. The

		phone variant is still undecided
Image upload as multipart form data	Base64 JSON, image URL	No size inflation, and URLs would need outbound requests and would imply the image had been stored somewhere
Rule A for the label, grouped sums for fusion evidence	Summing probabilities for everything	Summing over-predicted distress by 6 to 8 points because of the calibration change. Fusion still needs numbers, so I use sums only when a second signal exists
Decision-level fusion with weighted average	Feature-level fusion, a learned fusion model	Simple to inspect, and no paired training data exists to learn a fusion model
No default fusion parameters	A default face weight of 0.5	The decision region is between 0.50 and 0.60, so a default of 0.5 would silently decide the outcome
Sinhala-only text model with non-Sinhala rejected	Applying the model to English	It predicted calm for all 60 English test records, with no warning
Three application mood states	Direct sentiment polarity	Polarity is not the same axis as mood. Negative wording is not necessarily

		distress

Table 4.4: Design decisions and alternatives.

4.4 Detailed design

4.4.1 Facial expression model

Item	Value	Item
Base network	MobileNetV2, alpha 1.0, ImageNet weights, no top, average pooling	Base network
Head	Dropout 0.2, then Dense with 7 outputs and softmax	Head
Input	96×96×3, float32	Input
Parameters	2,266,951 total, 1,848,583 trainable	Parameters
Unfreezing	Deep: from about 64.5 percent of the network onward, with batch normalisation frozen	Unfreezing
Output	Seven calibrated probabilities, in the order angry, disgust, fear, happy, neutral, sad, surprise	Output
Calibration	Temperature 5.727, built into the exported graph	Calibration

Table 4.5: Facial model design.

4.4.2 Facial service contract

Rule	Detail
Accepted input	JPEG, PNG, BMP or WEBP, at most 8 MB, at least 16 pixels on each side
Response	Model version, model hash, class order, seven probabilities, predicted class, confidence, calibrated flag, label space

Errors	One envelope with a stable error code (for example `invalid_image`, `image_too_large`, `model_load_failed`)
Contract violation	If the loaded model's tensor shapes or hash do not match the contract, the service refuses to serve and returns `contract_violation`
Privacy	No image storage. Error responses never reveal file paths or training details
Versioning	Retraining, a new file, a quantisation change or a preprocessing change is a major version bump, because stored predictions then stop being comparable

Table 4.6: Facial service rules.

4.4.3 Sinhala sentiment model and service

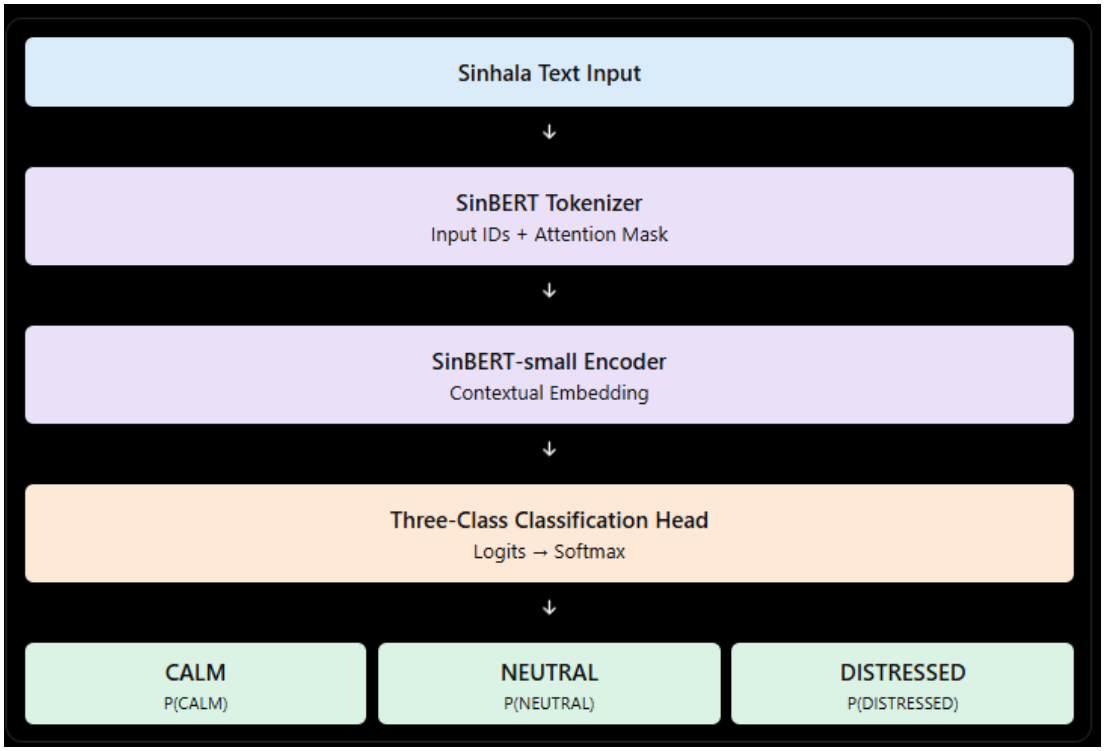


Figure 4.2: Sinhala sentiment model architecture

4.4.4 Fusion layer

Mapping seven expressions to three states.

Facial expression	Mood state
happy	CALM
neutral, surprise	NEUTRAL
angry, disgust, fear, sad	DISTRESSED

Table 4.7: Frozen seven-to-three mapping.

Two outputs from the face. For a standalone prediction, the label is the translation of the single most probable expression (Rule A). For fusion, the face supplies three grouped scores: calm equals happy, neutral equals neutral plus surprise, distressed equals angry plus disgust plus fear plus sad. The layer also carries Rule A's label and the model's top probability as a confidence value.

Combination rule. For each state, the final score is the face weight times the face score plus the text weight times the text score, with the two weights summing to 1. The state with the highest final score wins.

Availability cases.

Face	Text	Result
Available	Available	Weighted combination
Unavailable	Available	Text scores pass through unchanged
Available	Unavailable	Face label from Rule A, never the grouped sums
Unavailable	Unavailable	

Table 4.8: Fusion behaviour by availability.

4.4.5 Chatbot behaviour (intended design)

The chatbot is the consumer of the mood state, and it is not part of the ML work completed for this thesis. The intended behaviour is that a calm or neutral state leaves its tone unchanged, a distressed state makes replies gentler and shorter, and

sustained distress suggests she speak to her midwife or doctor. A safety signal, such as a message about self-harm, is a different concept from mood. A precedence rule for it has been proposed but not implemented, so I make no claim about it.

4.5 Standards and professional practice

Area	Practice followed
Referencing	IEEE style, as the template recommends
Data integrity	SHA-256 hashes for datasets, splits, the locked test file and model files
Experimental discipline	Validation for selection, one-time use of test sets, a measured noise floor, and a run record for every experiment
Versioning	Semantic versioning of the face model and service [25]
Interfaces	REST over HTTP with multipart upload and JSON responses
Text handling	Unicode for Sinhala, checked for byte-for-byte survival
Face detection	MediaPipe BlazeFace, a published mobile face detector [23]
Data protection	Sri Lanka's Personal Data Protection Act No. 9 of 2022 applies to health-related personal data [24]. I handle no live user data in this thesis, but any deployment would need to comply
Ethics	No participant data was collected. SLIIT ethical review is required before any user study
Responsible AI	No diagnostic claims, bias limits stated, missing evidence never invented, results reported even when they miss targets

Table 4.9: Standards and professional practice.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

I did the work in four separate environments, each chosen for one job. I kept them separate so that a result could be reproduced on a different machine.

Purpose	Environment	Main tools
Dataset audit and detector validation	Conda environment (Python 3.10), Windows	MediaPipe (BlazeFace), OpenCV, Pillow, pandas
Facial model training and evaluation	WSL2 Ubuntu on a laptop with an NVIDIA RTX 3050 (6 GB)	TensorFlow 2.21.0, Keras 3.15.1, mixed precision (float16 compute, float32 output layer)
Model conversion and benchmarking	WSL2 Ubuntu (Python 3.11)	TensorFlow Lite converter, XNNPACK delegate
Sinhala sentiment	CPU only, Windows	Hugging Face Transformers, PyTorch, safetensors checkpoints

Table 5.1: Development environments.

5.2 Component implementation

5.2.1 Data audit and cleaning

I audited all 35,887 images three ways: exact byte hashes (MD5), a perceptual hash with an exact match, and a perceptual hash with a Hamming distance of at most 3. I also measured pixel statistics and quality anomalies (12 fully black images, 14 near-constant ones and 67 brightness outliers). The cleaning step applied three rules to the training split only: drop cross-split duplicates (703 files), drop within-train duplicate redundancy (1,064 files), and drop anomaly-flagged images (53 files), 1,808 distinct files in total. I implemented cleaning as manifest files, with the raw data left alone.

5.2.2 Facial model training

I trained in two stages. The first stage cached frozen-backbone embeddings, which made it fast enough to repeat over three seeds and measure the noise floor. The second stage fine-tuned in seven runs, changing one factor at a time and carrying the winner forward.

Stage	Factor	Options	Chosen
A	Augmentation	Off, on	Off
B	Unfreeze depth	Shallow (20 tail layers), deep (from 64.5%)	Deep
C	Batch normalisation	Frozen, trainable	Frozen
D	Class weighting	Unweighted, balanced	Unweighted
E	Seed repeat	42, 1337, 2024	Confirms result

5.2.3 Calibration and conversion

I fitted temperature scaling on validation by recovering the logits from the saved probabilities and dividing them by a single temperature. The fitted value was 5.727. I then exported four TensorFlow Lite variants from the calibrated model: float32, float16, dynamic int8 and full-integer int8. The full-integer variant was rejected (Chapter 6).

5.2.4 Facial inference service

I wrote the service from the contract described in 4.4.2. It checks the model file's hash and tensor shapes at startup, and refuses to serve if either differs. It validates every upload (format, size, minimum dimensions) and returns the stable error codes. It preprocesses with Pillow, following the five steps in 4.4.1. Detail strings go to the server log and are not returned to callers unless a debug flag is set.

5.2.5 Sinhala sentiment model and service

I ran Experiment 01 (90 records) and Experiment 02 (299 records) using the settings in 3.3.2. A dedicated evaluation script, not a notebook, ran the one authorised test, and it checked the locked file's identifiers and hash before running inference. It saved identifiers, labels and probabilities only, not the text. The web service wraps the chosen checkpoint, rejects nothing itself, and logs no user text.

5.2.6 Fusion layer

I implemented the rules in 4.4.4 as a module with no default parameters. It refuses to start unless it receives explicit values for all of them. It discards text whose language is not Sinhala, and it returns the signals used alongside the mood state, so downstream code always knows what the answer rested on.

5.2.7 Problems I met

Problem	**Cause**	**What I did**
A saved model failed to load with an error about an unexpected argument	The model was saved by Keras 3.15.1 and loaded under 3.13.0. Newer Keras writes configuration keys that older versions cannot read	Loaded under the matching version
The calibrated model could not be loaded even under the same version	The temperature-scaling wrapper does not survive Keras serialisation	The calibrated model exists only as the four exported TensorFlow Lite files. To re-derive it, I would refit the temperature from the base model
A correlation was reported as NaN	One test image had no face detection, and the function propagates NaN	Recomputed with NaN handling
Latency looked worst for float16	x86 processors have no native half-precision	Treated the figure as a development-machine

	compute, so the converter inserts dequantisation steps	result only, not a phone result
The TensorFlow Lite interpreter warns it will be removed	Deprecation	Noted a migration path to a newer runtime

Table 5.3: Implementation problems and responses.

5.3 Integration with the other components

The integration between my parts is real. The integration with the rest of MaternaLink is only designed.

Between my own parts. BlazeFace feeds the facial service, the services feed the fusion layer, and the fusion layer returns a mood state. I tested the facial service and sentiment service separately and the fusion layer against its specification. I did not run a single end-to-end session with a real camera and a real message, because the backend that would connect them is not built.

With the backend. The integration guide I wrote for the backend developer sets out how the services should be called:

Setting	**Value**
Connection timeout	10 seconds
Read timeout, normal	15 seconds
Read timeout, after a cold start	75 seconds
Total budget including a retry	about 90 seconds

Table 5.4: Recommended timeouts for the face service.

5.4 Verification and software testing

I tested what the evidence lets me test: that each part reproduces its recorded results and follows its specification. I could not test how accurate the combined system is.

What was checked	Result
Preprocessing in the service against the training pipeline	Difference of exactly 0.0 across about 99 million values
Deployed model files against the certified ones	Hashes match
Temperature scaling changes any prediction	0 of 3,589 predictions changed
TensorFlow Lite float32 against the Keras model	99.5% prediction agreement on a 200-sample subset
Sentiment code against the recorded results, original batching	Difference of 1.11e-16, 76 of 76 predictions match
Sentiment code, one message at a time	Difference of 4.47e-07, 76 of 76 match
Sentiment web service against direct code	Difference of exactly 0.0 across 228 values, 76 of 76 match
Sinhala text through the web layer	Survived byte for byte
User text stored or logged	None, across 85 requests
Every documented error code in both services	Tested
Fusion against its specification	15 of 15 registered checks pass (16 tests including the sweep)
Missing-signal cases and the four “unknown” conditions	All tested

Fusion refuses to run without parameters	Tested
Non-Sinhala text rejected at several confidence levels	Tested
Face-only input returns the Rule A label at every weight	Tested

Table 5.5: Verification results.

5.5 Deployment and operation

The two services run on a development machine. I planned a Hugging Face model repository and Space for the facial service, with the float32 variant, because Hugging Face runs on x86 where float32 was the most accurate and fastest. I did not upload anything. The sentiment checkpoint has not been uploaded anywhere either.

The phone variant is still open. Notebook 07 recommended float16, but notebook 08 found that float16 had the largest memory footprint on the development machine (about 18.5 MB of tensor buffers against 14.0 MB for float32 and 7.8 MB for dynamic int8). That is a proxy measured on x86 and it may reverse on a phone with native half-precision support. The choice depends on two things I have not done: a phone benchmark, and a test of whether temporal smoothing absorbs the 8.5 percent of predictions that dynamic int8 changes.

Phone benchmark. No Android device was connected, so I recorded latency as unmeasured and wrote an executable protocol instead: four variants, one, two and four threads, XNNPACK on and off, at least 50 warm-up and 200 timed runs, and a thermal throttling check across repeated batches.

Operation. The services return a model version and file hash with every prediction, so any stored result can be traced to the exact model. Any retraining or quantisation change counts as a major version and invalidates stored predictions for comparison. Any retraining also invalidates the calibration, which must be refitted.

Backing up. Both test sets are now spent, so neither model could be honestly regenerated if its files were lost. I backed up the model weights and checked them against their hashes.

6 RESULTS AND VALIDATION

6.1 Research results

6.1.1 Facial expression model: validation results

Fine-tuning gave a large, real improvement over the frozen backbone.

Stage	Validation macro-F1	Validation accuracy
Frozen MobileNetV2 backbone	0.4357 ± 0.0061	0.4791 ± 0.0035
Fine-tuned MobileNetV2	**0.6063 ± 0.0054**	**0.6267 ± 0.0067**
Improvement	**+0.1706**	+0.1476

Table 6.1: Validation results, mean and standard deviation over three seeds.

6.1.2 Facial expression model: held-out test results

Model	Test accuracy	Test macro-F1	Test weighted F1
Frozen baseline	0.4798	0.4142	0.4708
Fine-tuned	**0.6289**	**0.6036**	0.6262
Improvement	+0.1491	**+0.1894**	

Table 6.2: Test results (3,589 images, used once).

The test improvement (+0.1894) is larger than the validation improvement. The fine-tuned model's validation score fell by only 0.0085 on test (0.6121 on validation for the same seed, 0.6036 on test), barely above the noise floor. The frozen model's gap was 0.0261. I conclude that selecting on validation did not overfit it.

Removing the 8 flagged anomaly images from the test set changed accuracy by +0.0008 and macro-F1 by +0.0008, so junk images do not meaningfully depress the result.

Class	Recall	Precision	F1	Support
happy	0.877	0.813	0.844	879

surprise	0.700	0.770	0.733	416
neutral	0.626	0.573	0.599	626
disgust	0.491	0.675	0.568	55
angry	0.497	0.551	0.523	491
sad	0.471	0.498	0.484	594
fear	0.477	0.472	0.475	528

Table 6.3: Per-class test results for the fine-tuned model.

Happy and surprise separate cleanly. The four negative expressions (angry, disgust, fear, sad) score between 0.47 and 0.50 recall and are confused mainly with each other. disgust reached F1 0.568 from only 351 training images, so class confusability, not class imbalance, appears to be the binding limit.

Of the three confusion predictions I made from looking at the images, two held and one did not:

Prediction	**Outcome**	**Evidence**
Sad and neutral are confused	Confirmed	sad→neutral 0.19, neutral→sad 0.12
Disgust is confused with angry	Confirmed	disgust→angry 0.27, the largest off-diagonal rate
Fear and surprise are confused	Not the main confusion	fear→surprise only 0.08

Table 6.4: Predicted confusions against measured ones.

The dominant fear confusion was with sad (fear→sad 0.18, sad→fear 0.15), a pairing I had not predicted. What looks confusable to a person at 48×48 pixels is not necessarily what the network confuses.

6.1.3 Calibration

	Before	**After**

Expected Calibration Error (validation)	0.3010	**0.0126**
Fitted temperature	n/a	5.727
Predictions changed	n/a	**0 of 3,589**

Table 6.5: Effect of temperature scaling.

Before calibration, the model was severely overconfident. On the test set, 79 percent of predictions were made at a stated confidence of about 0.99, but only about 70 percent of them were correct. Predictions stated at about 85 percent were right 41 percent of the time. Even so, confidence carried useful signal: its AUROC as a correctness detector was 0.755 (0.5 would be useless). After scaling, the reliability curve lies close to the diagonal, and because no prediction changed, every accuracy and F1 figure above stays valid.

6.1.4 Where the model is weak

Check	**Result**
Darkest tenth of test images	Accuracy 0.599, macro-F1 **0.457**
Brightest tenth	Accuracy 0.699, macro-F1 **0.644**
Low pose or occlusion confidence	Accuracy 0.601
High pose or occlusion confidence	Accuracy 0.657
Brightness used as a shortcut	No. 51.5% of errors fall closer to the predicted class's mean brightness, against 50% by chance
Where the model looks (Grad-CAM)	The centre of the face, never the edges or watermarks
Attention on correct against wrong predictions	Nearly identical

Table 6.6: Robustness and validity checks.

Poor lighting costs 10 points of accuracy but 19 points of macro-F1, so it hurts rare classes far more than accuracy suggests. This matters for use in dim homes. The model also shows no sign of exploiting brightness or watermarks, and its errors look like real expression confusions, not attention failures. One limit applies to the attention maps: the final feature map is only 3×3, so these maps support only "centre against edge" and "upper against lower" conclusions and nothing finer.

6.1.5 Three-state result and the miss rate

After applying the frozen mapping in Table 4.7, the test results were:

Metric	**Value**
Macro-F1	0.7681
Accuracy	0.7668
Balanced accuracy	0.7745
Weighted F1	0.7664

State	**Precision**	**Recall**	**F1**
CALM	0.814	0.878	0.845
NEUTRAL	0.674	0.688	0.681
DISTRESSED	0.801	0.757	0.778

Table 6.7: Three-state test results.

This is not an improvement over 0.6036. The model is unchanged. The mapping merges the four expressions that the model confuses most into one state, so a mistake between them now counts as correct. The two figures measure different tasks and should never be compared as progress.

Outcome	Count	Share
Correctly called DISTRESSED	1,263	75.7%
Called NEUTRAL	308	18.5%
Called CALM	97	5.8%
Total missed	**405**	**24.3%**

Table 6.8: What happened to distressed faces.

Roughly one in four distressed expressions was not flagged. The model predicts distress for 43.9 percent of images when the true share is 46.5 percent, so it under-calls it overall. This is the main reason a second signal is needed.

6.1.6 Sinhala sentiment model

Experiment	Records	Accuracy	Macro-F1	Note
01 (validation)	18	0.5556	0.3872	CALM recall and F1 were zero (2 CALM records)
02 (validation)	76	0.6842	0.6249	Selected checkpoint (epoch 5)
02 (locked test)	**120**	**0.3750**	**0.3620**	One evaluation

Table 6.9: Sinhala mood model results.

Experiment 02's per-class validation F1 was CALM 0.516, NEUTRAL 0.778 and DISTRESSED 0.581. The locked test result is only slightly above the chance level for three balanced classes (macro-F1 about 0.333).

Class	Precision	Recall	F1	Support
CALM	0.343	0.575	0.430	40
NEUTRAL	0.450	0.225	0.300	40

DISTRESSED	0.394	0.325	0.356	40

Table 6.10: Locked test results, English and Sinhala combined.

Confusion matrix (rows true, columns predicted): CALM [23, 6, 11], NEUTRAL [22, 9, 9], DISTRESSED [22, 5, 13].

The English problem. The test set holds 60 Sinhala and 60 English records, and the model was trained on Sinhala only. According to the supervisor findings in my records, it predicted CALM for all 60 English records. If that is correct, the matrix above splits into an English part and a Sinhala part, which I derived myself:

Subset	**Accuracy**	**Macro-F1**	**Note**
English (60)	0.333	0.1667	Every record predicted CALM, which is exactly chance
Sinhala only (60)	0.417	0.388	CALM recall 0.15, the worst class

Table 6.11: Test results by language (derived from Table 6.10's matrix and the all-CALM English finding).

This changes how the combined figures should be read. The high CALM recall of 0.575 in Table 6.10 is inflated by English records collapsing to CALM, and it is not evidence that the model detects calm well. On Sinhala alone, calm is the weakest class.

Why validation and test disagree. The validation score of 0.6249 fell to 0.362 on the combined test, but the language mix does not explain all of it. Sinhala-only test macro-F1 (0.388) is still far below the validation score. Plausible causes are the small validation set (76 records, 12 of them DISTRESSED), the way labels were produced, and a shift between the development records and the locked test records. My evidence cannot separate these. I therefore treat the model as a development model with weak standalone reliability, not as a validated one.

6.1.7 Fusion

Fusion has no accuracy, F1 or confusion matrix, for the reason given in 3.5. It passed all 15 registered conformance checks. The weight sweep gave the following picture:

Scenario type	**Outcome across face weights from 0 to 1**
Face and text agree (4 of 8 scenarios)	The answer never changes
Face and text disagree	The answer flips, and every flip falls between a face weight of 0.50 and 0.60

Table 6.12: Weight sensitivity.

The most important case is a face that reads NEUTRAL with text that reads DISTRESSED. Below a face weight of 0.60 the result is DISTRESSED, so the missed distress is recovered. At 0.60 or above, it is NEUTRAL and the miss reaches the user. An "equal weights" default of 0.5 sits right next to that boundary, so it is not a safe placeholder.

6.2 Product and technical results

6.2.1 Deployment variants

Variant	**Size (MB)**	**Macro-F1 (validation)**	**Change from Keras**	**ECE**	**Prediction agreement**
Keras, calibrated	24.5	0.6117	n/a	0.0126	n/a
float32	8.96	0.6122	+0.0005	0.0099	99.5%
float16	4.58	0.6090	−0.0027	0.0135	99.0%
dynamic int8	2.57	0.6071	−0.0046	0.0103	**91.5%**

Table 6.13: TensorFlow Lite variants (validation data; agreement on a 200-sample subset).

Float32, float16 and dynamic int8 are all within the noise floor on macro-F1. Dynamic int8's score looks fine, but it changes about one prediction in 12, a

difference the aggregate score hides. Full-integer int8 lost 18 times the noise floor, wrecked the calibration, and was larger than dynamic int8, so I rejected it. Its cause is likely per-tensor activation quantisation, a known failure mode for MobileNetV2's depthwise convolutions.

Memory gave an unexpected result. On the development machine, float16 had the largest memory footprint (18.5 MB of tensor buffers against 14.0 MB for float32 and 7.8 MB for dynamic int8), because the runtime adds dequantisation steps that rebuild float32 copies of the weights. This is an x86 proxy measurement and it may reverse on a phone with native half-precision support, so I do not report it as a phone figure.

Latency on the development machine (float32 1.78 ms, dynamic int8 2.25 ms, float16 2.62 ms, full int8 0.93 ms mean) is likewise not a phone result. **Phone latency was not measured.**

6.2.2 Service results

Result	**Value**
Preprocessing parity (service against training pipeline)	Exactly 0.0 difference over about 99 million values
Sentiment code against recorded results	Max difference 1.11e-16, 76 of 76 predictions match
Sentiment web service against direct code	Difference exactly 0.0, 76 of 76 match
User text stored or logged	None across 85 requests
Fusion conformance	15 of 15 checks pass

Table 6.14: Technical results.

6.3 User or stakeholder validation

I have no user or stakeholder validation to report. My proposal planned a survey of pregnant women, a usability test with 2 volunteers, and measures such as agreement with self-reported mood (target 70 percent) and satisfaction with the chatbot's tone (target 4 out of 5). I did not carry these out. They needed ethics

clearance, the finished app and participants, and none of those were in place for this thesis.

The only human-based evidence I have is annotator agreement. Two annotators agreed on 91.67 percent of the 180 non-frozen development records and on 93.33 percent of the 90 Sinhala records, with disagreements adjudicated. This says something about the clarity of the labelling guideline. It says nothing about whether women find the system useful or whether the labels reflect how they feel.

6.4 Integration results

Each of my parts works and was verified on its own (Table 6.14 and Table 5.5). I did not run a complete end-to-end session with a real camera, a real message and a chatbot reply, because the backend and chatbot that would join them are not built. I therefore have no measured end-to-end latency, no tested behaviour when the face service is cold-starting, and no result from the doctor-facing summary. The backend integration guide specifies how these should behave (Table 5.4), but specification is not a result.

6.5 Summary of findings

Hypothesis	**Outcome**	**Evidence**
H1a: test accuracy at least 65%	**Not supported for seven classes; met for three states, with a caveat**	Seven-class accuracy 0.6289 is just under 65%. The three-state accuracy of 0.7668 clears it, but that task is easier, not better
H1b: fine-tuning beats the baseline by more than the noise floor	**Supported**	+0.1706 on validation, 26 times the noise floor, confirmed on test (+0.1894)
H2: calibration error	**Supported**	ECE 0.0126, 0 of 3,589

below 0.1 with no prediction changed		predictions changed
H3: a compressed variant stays within noise and agrees closely	**Supported**	float32 and float16 within noise, with 99.5% and 99.0% agreement. Dynamic int8 fails on agreement (91.5%), and full int8 fails on both
H4: Sinhala test accuracy at least 70%	**Not supported**	Accuracy 0.375 on the combined test and 0.417 on Sinhala alone
H5a: fusion follows its specification	**Supported**	15 of 15 conformance checks
H5b: the outcome depends on the face weight when signals disagree	**Supported**	Every flip lies between 0.50 and 0.6

Table 6.15: Hypotheses and outcomes.

7 DISCUSSION

7.1 Interpretation of findings

The facial model works, within limits. Fine-tuning lifted test macro-F1 from 0.4142 to 0.6036, and the gain was larger on test than on validation. I read this as evidence that the frozen backbone was the bottleneck. ImageNet features were never adapted to 48×48 grayscale faces, and letting the network adjust its internal layers fixed that. The small gap between validation and test (0.0085) suggests my selection process was sound and that I had not quietly overfit the validation set.

The remaining errors look like real confusion, not a fixable bug. Three results point the same way. The attention maps showed the model looking at the face, not at the corners or watermarks. Errors showed the same attention pattern as correct predictions. And brightness did not act as a shortcut. So the model is learning expression, and it fails where expressions really are hard to tell apart. The four negative classes (angry, disgust, fear, sad) all sit near 0.47 to 0.50 recall and are confused mostly with each other. Since disgust did better than expected from only 351 training images, I read the limit as class confusability, not class imbalance.

The 0.7681 figure should not be celebrated. The three-state score is higher because the mapping merges the confusable classes, not because the model improved. What matters for a wellbeing tool is that 24.3 percent of distressed faces were missed. A reader who sees only 0.7681 would draw the wrong conclusion, so I put the miss rate next to it.

Calibration mattered more than I expected. A model that claimed 99 percent confidence while being right about 70 percent of the time would have corrupted any weighted combination, because fusion treats the numbers as meaningful. One scalar fixed this with no change to any prediction. I would now treat calibration as a required step for any model whose confidence feeds another component.

The Sinhala model is not reliable. I do not claim it detects mood. The test score was close to chance, the English records collapsed to a single confident class, and the validation score was too optimistic to trust. The most likely reasons are the small

amount of data, a mood label that is hard to apply consistently, and the fact that SinBERT was pretrained and evaluated on polarity tasks like news comments, which differ from conversational pregnancy messages. I cannot separate these causes with my evidence.

Fusion is a design, not a result. Its value lies in the missing-signal rules and in the weight finding. The most useful result is that disagreement between face and text is decided in a narrow band of face weights between 0.50 and 0.60, which is exactly where a default of 0.5 would sit.

7.2 Comparison with literature baselines and existing solutions

Facial expression recognition. The FER-2013 authors reported human accuracy of about 65 percent [13], and published state-of-the-art models reach about 73 percent according to my notebook records. My test accuracy of 0.6289 sits just below the human figure and about ten points below the state of the art. The comparison needs three cautions:

- My model is a compact MobileNetV2 meant for mobile use. Published top scores generally come from much larger models and extra tricks.
- I removed duplicates and cross-split leakage from training, and I used the test set once. Some published numbers come from protocols that tuned on the test set or kept the duplicates, so a lower honest number is not necessarily a worse model.
- FER-2013 has label noise. I found 26 groups of byte-identical images filed under different emotions, which caps accuracy below 100 percent for any model.

I did not re-implement other methods under my protocol, so I cannot claim to be better or worse than them.

Calibration. Modern networks are known to be overconfident, and temperature scaling is the standard fix [17]. My result agrees with that literature. The fitted

temperature of 5.727 is large, which shows how overconfident the fine-tuned model had become.

Sinhala language models. SinBERT has been reported to outperform multilingual baselines on Sinhala classification [18], but on news-comment benchmarks. My finding that it performs poorly on a pregnancy mood task does not contradict that work. It supports the caution that those scores guide model choice and do not predict performance on a new domain.

Existing solutions. Table 2.1 listed what is available. My component does not replace any of them. Chatbots such as Woebot have shown benefit in a randomised trial [20], and my work has no comparable evidence. Where mine differs is in design: it targets Sinhala, it senses mood passively, and it states what it does when a signal is missing. Whether this design is better for users is untested.

Multimodal fusion. Reviews say that combining modalities usually helps because each fails in different conditions [7], [19]. The face model's miss rate and the text model's weakness make this argument fit my case in principle. But those reviews rely on datasets in which several signals are recorded from the same person, and I have no such data, so I cannot show that fusion helps.

7.3 Research contribution

I see four contributions, from strongest to weakest evidence.

1. **A leakage-aware evaluation of FER-2013 for a mobile model.** I measured duplicates and leakage, kept the canonical protocol, measured a noise floor, and used the test set once. Anyone reusing the manifests can reproduce the numbers.
2. **A worked example of why single numbers mislead.** Merging classes raised macro-F1 by 0.16 with no model change. Confidence of 99 percent meant 70 percent accuracy. A quantised model kept its macro-F1 but changed one prediction in 12. A combined test score hid a total failure on English. I think this is the most transferable lesson in the thesis.

3. **A fusion design that refuses to guess.** It has no default parameters, returns "unknown" instead of inventing a signal, and rejects text in an unsupported language. These choices are small, but they address failures that I observed.
4. **A negative result on Sinhala mood detection.** I show that a small annotated set and a polarity-pretrained model are not enough for this task. A negative result with a clear protocol is useful to anyone planning the same work.

I make no claim that the system improves maternal outcomes or detects any condition.

7.4 Product contribution

I produced three working software parts: a facial inference service with a written contract and integrity checks, a Sinhala sentiment service verified against its recorded results, and a fusion layer with specification tests. I also produced a backend integration guide, so another developer can use the services without reading my notebooks. The services are not deployed, there is no backend, and the chatbot that would use the mood state is not part of what I built. The product contribution is therefore a verified core, not a finished application.

7.5 Limitations and threats to validity

Internal validity (are my conclusions true of my own experiments?)

- Several choices were made on validation only, and the Sinhala validation set is small (76 records, 12 of them distressed), so its scores are noisy.
- I fixed the three-state mapping by supervisor judgement, and the data could not settle it. A different mapping would give a different three-state score.
- The temperature was fitted on validation. I assume it carries over to test, but checking would have needed a second test touch.
- The dynamic int8 parity check used only 200 samples.

Construct validity (do my measures capture what I claim?)

- A facial expression is not an emotion, and a message is not a mind. The system estimates appearance and wording.
- My three mood states are application constructs defined by my guideline. A message was labelled distressed if distress was expressed or implied, which is not the same as a person being distressed.
- FER-2013 expressions were collected from posed and web images, and they may differ from the spontaneous expressions of a woman chatting at home.

External validity (do the results carry over?)

- **Demographics.** FER-2013 is heavily Western and has few South Asian faces. I never measured performance for Sri Lankan users, and this is the most serious limitation of the project.
- **Lighting and cameras.** Accuracy dropped in dim images, and phone cameras differ from the web images in FER-2013.
- **Language.** The text model covers Sinhala only, and the data was small.
- **Population.** No pregnant woman took part. Nothing here is known about how pregnancy affects expression or wording.

Data and provenance. Your provenance record shows that the original candidate pool contains authoring metadata, and that part of the Sinhala data was supplied as annotated “demo” records. I have to state clearly where the messages came from, since messages written for a study do not behave like messages from real users.

Conclusion validity.

- Two of seven hypotheses were not supported. I did not adjust a hypothesis after seeing its result.
- The per-language decomposition of the text result is derived from the combined matrix and needs confirming against the saved predictions.
- Fusion has no accuracy figure, so no conclusion about its benefit is justified.

Things I did not measure. Phone latency and memory, behaviour on Linux for the sentiment service, end-to-end behaviour, real-user outcomes, and any clinical effect.

7.6 Implications

For the MaternaLink project. The face signal alone should not be treated as a dependable distress detector. A design that depends on it would miss about a quarter of distressed faces. The app should present mood estimates as gentle prompts, never as findings, and should keep working without the camera. The group should choose the face weight with evidence, not by default, because the choice decides whether a missed face signal can be rescued by text.

For research. The most valuable next step is data, not a bigger model: a face dataset with South Asian faces, a larger and better annotated Sinhala set, and a paired dataset in which face and message come from the same person at the same moment with a verified mood. Without the paired set, fusion cannot be measured for any group.

For practice. Anyone building a similar tool should report per-class results, measure calibration, check prediction agreement after compression, and test what happens when the input is in the wrong language. All four of these caught something in my project that a headline number hid.

For ethics. A system that guesses at a pregnant woman's mood carries a risk in both directions: missed distress, and false alarms that frighten someone. Any pilot should be supervised by clinicians, should state what the system cannot do, and should go through ethical review before it touches a participant.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis set out to build and honestly evaluate the mood detection pipeline of a bilingual pregnancy support chatbot for Sri Lanka. It had three parts: a facial expression model, a Sinhala text model, and a layer that combines them.

The facial part worked best. A fine-tuned MobileNetV2 reached 0.6036 macro-F1 and 0.6289 accuracy on seven expressions, evaluated once on held-out data after I removed duplicates and leakage from the training set. Temperature scaling reduced its calibration error from 0.30 to 0.013 without changing a single prediction. Mapped to three application states, it reached 0.7681 macro-F1, but that reflects an easier task and not a better model. The figure that matters is that 24.3 percent of distressed faces were missed.

The Sinhala text model did not meet its target. It scored 0.6249 macro-F1 on validation and 0.3620 on the locked test set, close to chance, and it fails silently on English. I treat it as a development model, not a validated one.

The fusion layer follows its specification, handles a missing camera, and returns "unknown" instead of guessing. I could not measure its accuracy because no dataset pairs a face and a message with a verified mood. The one strong finding is that when face and text disagree, the outcome depends on a face weight in a narrow band between 0.50 and 0.60.

My overall conclusion is that a small, calibrated facial model is feasible for this application, but neither signal is yet reliable enough to detect distress on its own. The system should not be used for any clinical purpose. The main value of this work is a transparent, reproducible evaluation that shows where the evidence stops.

8.2 Achievement of objectives

Objective	**Outcome**	**Evidence**
01. Build a facial	**Achieved, with the**	Fine-tuning beat the baseline

expression model that is accurate enough to be useful and small enough for a phone	**accuracy target narrowly missed**	by 26 times the noise floor. Seven-class test accuracy was 0.6289 against a 65% target. The three-state accuracy of 0.7668 clears it, on an easier task
02. Make the model's confidence trustworthy	**Achieved**	ECE fell from 0.3010 to 0.0126, with 0 of 3,589 predictions changed
03. Package the model for deployment	**Partly achieved**	Four variants were exported and compared. Full int8 was rejected, and float32 and float16 were within noise. Phone speed was **not measured**, and the phone variant is undecided
04. Build a Sinhala text model for calm, neutral and distressed	**Built, but the target was not met**	Locked test accuracy was 0.375 against a 70% target. English was not supported
05. Design a fusion layer	**Achieved for design and specification, not for accuracy**	15 of 15 checks passed. Accuracy was not measurable and the parameters were not chosen
06. Evaluate honestly and state what the evidence does not show	**Achieved**	Each test set was used once, every miss is reported, and Section 7.5 lists the limits

Table 8.1: Achievement of objectives.

8.3 Individual research and product contributions

Research contributions.

- A leakage-aware evaluation of FER-2013 for a compact model, with a measured noise floor and a single use of each test set.
- A demonstration that headline numbers can mislead: class merging, overconfidence, quantisation and mixed-language test sets each hid a problem that a finer measure revealed.
- A fusion design with explicit missing-signal behaviour, and a sensitivity result showing that the face weight decides the outcome only between 0.50 and 0.60.
- A negative result on Sinhala mood detection with a documented protocol.

Product contributions.

- A facial inference service with a written contract, a model integrity check and stable error codes.
- A Sinhala sentiment service verified to reproduce its recorded results.
- A fusion layer with no default parameters, and 15 conformance checks.
- A backend integration guide that lets another developer use the services without reading the notebooks.

8.4 Future directions

1. **Better face data.** FER-2013 has few South Asian faces, and performance for Sri Lankan users has never been measured. A suitably diverse dataset, collected or licensed with ethical approval, is the most important next step.
2. **A paired face-and-text dataset.** Collecting a person's face and message at the same moment with a verified mood label would make fusion measurable for the first time, and would let the six fusion parameters be chosen with evidence. This is the largest outstanding piece of work.
3. **A larger, better annotated Sinhala set.** The approved calibration exercise for annotators was prepared but never run. Running it, collecting more

records (especially CALM and borderline cases) and testing agreement would show whether the weak result is a data problem or a labelling problem.

4. **English sentiment and language detection.** English needs its own model and validation. Until then, a language detector must stop non-Sinhala text from reaching the Sinhala model. That detector does not exist yet.
5. **Improving distress detection.** Options include training specifically to separate the negative expressions, using higher-quality images, or accepting more false alarms in return for fewer misses. Each should be judged against the 24.3 percent miss rate.
6. **A phone benchmark and a variant choice.** Run the prepared protocol on a named Android device, then test whether temporal smoothing absorbs the 8.5 percent of predictions that dynamic int8 changes.
7. **Temporal smoothing.** Averaging over several frames is likely more stable than judging one, but its window must be tested.
8. **Backend, deployment and the chatbot.** Build the backend, deploy the services, connect the chatbot, and run a full end-to-end session. Run the sentiment service on Linux first, since it was only verified on Windows.
9. **A user study.** After ethical review, test with adult volunteers, then with pregnant women under clinical supervision. Measure agreement with self-reported mood and satisfaction with the chatbot's tone.
10. **A safety design.** Distress messages about self-harm need a precedence rule that is separate from mood. A precedence rule has been proposed but not built.
11. **Recalibration.** Any retraining invalidates the calibration, which must be refitted.

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GLOSSARY

Term	*Definition*
Argmax agreement	*The share of individual predictions that stay the same when a model is converted or compressed. It can reveal changes that an overall score such as macro-F1 hides.*
Balanced accuracy	*The average of the recall of each class, so every class counts equally regardless of its size.*
BlazeFace	*A small, fast face detector from MediaPipe. In this project it finds and crops the face before the facial model sees it.*
Calibration	*How well a model's stated confidence matches how often it is right. A model that says "90 percent" should be correct about 90 percent of the time.*
Class imbalance	*A dataset in which some classes have far more examples than others. Here the ratio of the largest to the smallest class was 20.01 to 1 after cleaning.*
Confidence	*The probability a model assigns to its chosen answer. A high value does not guarantee the answer is right.*
Confusion matrix	*A table showing, for each true class, how often the model predicted each class.*
Conformance testing	*Checking that software follows its written specification. It is not the same as checking that the software is accurate.*
Decision-level fusion	*Combining the outputs of separate models after each has made its own decision, instead of merging their raw inputs.*
Design Science Research	*A research approach in which the researcher builds an artefact, evaluates it against stated criteria and refines it.*
Fine-tuning	*Continuing to train a pretrained network on a new task so its internal layers adapt, not just its final layer.*
Frozen	*Fixed and not allowed to change. A frozen backbone has weights that are not updated. A frozen decision or test set must not be altered, so that a reported result stays valid.*
Fusion	*Combining several sources of evidence, here*

	face and text, into one answer.
Generalisation gap	*The drop in performance between data used to choose a model (validation) and new data (test). A small gap suggests the model was not overfit.*
Grad-CAM	*A technique that shows which parts of an image a network relied on for a prediction.*
Grouped probability sums	*Adding the probabilities of related expressions to give one score per mood state. Calm equals happy, neutral equals neutral plus surprise, and distressed equals angry plus disgust plus fear plus sad.*
Macro-F1	*A score that averages the F1 of each class equally, so a model cannot score well by ignoring rare classes. It runs from 0 (worst) to 1 (best).*
Missing modality	*A signal that is not available, for example when the user turns off the camera.*
Modality	*One source of information. This system has two: the face and the text.*
Mood state	*One of the three application categories used here: CALM, NEUTRAL or DISTRESSED. These are not medical diagnoses.*
Noise floor	*The natural variation in results between repeated training runs with different random seeds. Here it is 0.0065 macro-F1, and smaller differences are not treated as real.*
Overconfidence	*A model that states higher confidence than its accuracy justifies.*
Pretraining	*Training a network on a large general dataset before adapting it to a specific task.*
Quantisation	*Storing a model's numbers at lower precision to reduce its size and sometimes speed it up, at a possible cost in accuracy.*
Rule A	*Taking the facial model's single most probable expression and translating that one label into a mood state.*
Seed	*A number that sets a program's random choices, so a run can be repeated exactly.*
Softmax	*The final step in a classifier that turns raw scores into probabilities that add up to 1.*
Temperature scaling	*A correction that divides a model's raw outputs by one number to fix overconfidence.*

	It does not change which class wins.
Test set	*Data held back and used once for the final measurement. Reusing it would make the result unreliable.*
Transfer learning	*Starting from a network pretrained on one task and adapting it to another, which needs less data than training from scratch.*
Unknown	*The fusion layer's output when it has no usable evidence. It is a system decision, never a model prediction.*
Validation set	*Data used to make choices during development, such as which settings to keep.*
Valence and arousal	*Two axes for describing emotion: how pleasant or unpleasant it is, and how calm or excited.*

APPENDICES

Appendix A Research instruments or extended results

This appendix holds supporting detail for Chapters 3 and 6. Chapter 6 cites it where needed.

A.1 FER-2013 class distribution

Class	Train	Train %	Validation	Test
angry	3,995	13.92	467	491
disgust	436	1.52	56	55
fear	4,097	14.27	496	528
happy	7,215	25.13	895	879
neutral	4,965	17.29	607	626
sad	4,830	16.82	653	594
surprise	3,171	11.05	415	416
Total	**28,709**	100	**3,589**	**3,589**

Table A.1: Original class counts. The ratio of happy to disgust in training is 16.55 to 1.

A.2 Effect of cleaning on the training set

Class	Original	Cleaned	Removed	Removed %
surprise	3,171	2,454	717	22.6
disgust	436	351	85	19.5
fear	4,097	3,803	294	7.2
angry	3,995	3,783	212	5.3
sad	4,830	4,672	158	3.3
neutral	4,965	4,814	151	3.0

happy	7,215	7,024	191	2.6

A.3 Duplicate and leakage audit

Method	**Duplicate groups**	**Files affected**	**Cross-split groups**	**Cross-split files**
MD5 (exact bytes)	1,516	3,369	557	1,323
Perceptual hash, exact	1,581	3,556	588	1,422
Perceptual hash, distance ≤ 3	2,051	5,100	807	2,311

A.4 BlazeFace calibration grid

Model selection	**Upscale**	**Input (px)**	**Detection rate**	**Mean confidence**
0	**1**	**48**	**0.9997**	**0.722**
0	2	96	0.9997	0.723
0	4	192	0.9997	0.724
1	1	48	0.8447	0.334
1	2	96	0.8713	0.331
1	4	192	0.8297	0.322

A.5 Validation results by class (fine-tuned against frozen)

Class	**Precision**	**Recall**	**F1**	**Support**	**Frozen**	**Change**

					recall	in recall
angry	0.513	0.514	0.513	467	0.334	+0.180
disgust	0.718	0.500	0.589	56	0.214	+0.286
fear	0.514	0.488	0.501	496	0.224	+0.264
happy	0.799	0.844	0.821	895	0.617	+0.227
neutral	0.534	0.591	0.561	607	0.572	+0.020
sad	0.556	0.505	0.530	653	0.412	+0.093
surprise	0.783	0.757	0.770	415	0.646	+0.111

A.6 Calibration before correction (test set)

Stated confidence	Empirical accuracy	Predictions
0.989	0.698	2,836 (79% of test)
0.851	0.413	235
0.754	0.362	163
0.651	0.374	171
0.546	0.328	128
0.462	0.245	49

A.7 Effect of a confidence threshold (test set, uncalibrated model)

Threshold	Coverage kept	Accuracy on kept	Correct predictions discarded
0.90	79.0%	0.698	277
0.95	72.6%	0.725	369

0.99	60.3%	0.768	594
0.999	46.7%	0.822	881

A.8 Latency on the development machine (not a phone result)

Variant	**Mean (ms)**	**95th percentile (ms)**
full int8	0.93	1.17
float32	1.78	2.58
dynamic int8	2.25	3.29
float16	2.62	5.30

A.9 Sinhala development data

Set	**CALM**	**NEUTRAL**	**DISTRESSED**	**Total**
Non-frozen pool, English	8	49	33	90
Non-frozen pool, Sinhala	9	44	37	90
Dev-v2 (Sinhala)	70	183	46	299
Dev-v2 train	52	137	34	223
Dev-v2 validation	18	46	12	76
Locked test (each language)	20	20	20	60

A.10 Sinhala model settings (Experiment 02)

Setting	**Value**
Checkpoint	SinBERT-small, pinned revision `7059f20a…acb6a`
New head	Three-class RoBERTa classification head
Maximum length, batch size	512 tokens, 8
Optimiser	AdamW, learning rate 2e-5, weight decay 0.01
Epochs, warm-up	At most 5, ratio 0.1
Class weights (training only)	CALM 1.4295, NEUTRAL 0.5426, DISTRESSED 2.1863
Selection	Validation macro-F1, early stopping patience 2
Split seed	20260828

Appendix B Academic integrity and AI use disclosure

Generative AI tools were used during this project to assist with drafting explanatory text, debugging code, and formatting this document. All model training, hyperparameter search results, and evaluation metrics reported in this thesis were executed and independently verified by the candidate; no result was accepted without direct verification. Representative uses included: drafting narrative explanations of already-implemented technical decisions; assistance diagnosing the two engineering issues described tables values get by using AI.